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Ecologists investigated changes in grassland communities on large islands off the coast of Scotland between 1975 and 2010. On each island, they used data from a number of sites to determine the change in mean species richness and the change in mean index of diversity.

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Table 1 shows plant species recorded at one site, on one island, in 1975.

Table 1

Species	Number of individuals
<i>Hydrocotyle vulgaris</i>	3
<i>Plantago maritima</i>	19
<i>Ranunculus acris</i>	3
<i>Hieracium pilosella</i>	3
<i>Calliargon cuspidatum</i>	10
<i>Prunella vulgaris</i>	16
<i>Pseudoscleropodium purum</i>	6

Calculate the index of diversity for this site using the formula:

$$d = \frac{N(N-1)}{\sum n(n-1)}$$

[2 marks]

$d =$ _____

Turn over for the next question

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